

Spatial Data and QGIS

Turning maps into variables you can put in a regression

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Monday, 21 September 2026

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Guest lecture · Empirical Development Economics · Undergraduate Economics, FEB UI

Can you see the crisis?



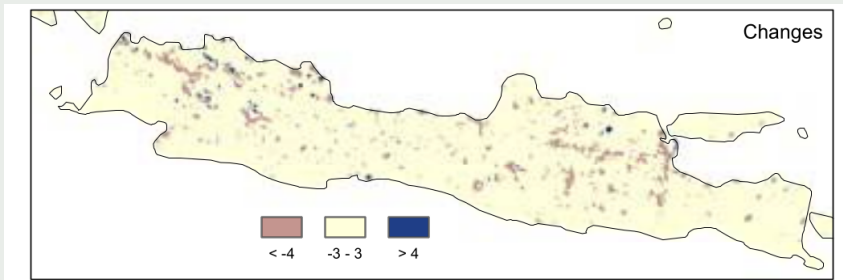
Java, two years in a row. The 1998 crisis happened in between.

The change is small. Average brightness in 1998 is only about **3 percent** lower, and city centres do not change at all because the sensor is already maxed out there.

Your eyes can barely catch it. **That is exactly why** we turn pictures into numbers.

Figure: Henderson, Storeygard & Weil (2012), *AER* 102(2), Figure 3 (part). [full reference](#) →

What your eyes miss



The third panel of the same figure: the *change* in light from 1997 to 1998, concentrated along the north coast city corridor.

Think about it: the authors did not trust their eyes either. They computed the difference first, then mapped it.

Figure: Henderson, Storeygard & Weil (2012), *AER* 102(2), Figure 3 (part). [full reference](#) →

Open these two before we start

The interactive lab

billytulungen.com/lab

Five things you can click while we talk. Runs in your browser, nothing to install.

Raster resolution · Projections and buffers · Map classification · Joining tables · Units of analysis

The practice files

billytulungen.com/lab → download

Twelve files. Unzip anywhere and open them straight from QGIS.

The README inside says which file each exercise needs. Everything is synthetic data built for this class.

Takeaway

You also need QGIS 3.x, the LTR release, from qgis.org. Please install it before the session starts.

Why Location Matters

Development questions always have a location

Infrastructure

Does a new road reduce poverty in a village?

You need the distance from every village to the road, year by year.

Natural resources

Do new plantations raise local wages?

You need yearly land cover change at the sub-district level.

Climate risk

How much rice output is lost to drought?

You need rainfall by district and by growing season.

Takeaway

All three have one thing in common: you have to build the key variable yourself, from a map. It is not published in any statistics book.

The first law of geography

“Everything is related to everything else, but near things are more related than distant things.”

Waldo Tobler, 1970

What this means for econometrics: observations that sit close together are not independent. We come back to this at the end.

From light to an economic variable



FIGURE 3. ASIAN FINANCIAL CRISIS: JAVA, INDONESIA

Dashed line: the GDP that was recorded. Solid line: GDP predicted from night lights alone.
Both drop together in 1998.

Think about it: Which kinds of places would a proxy like this be most useful for?

Figure: Henderson, Storeygard & Weil (2012), *AER* 102(2), Figure 3 (part). [full reference](#) →

Dots that became data: transmigration

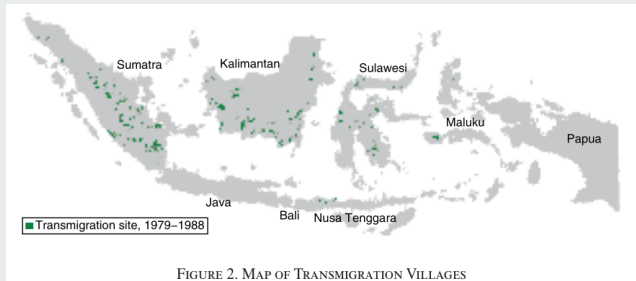


FIGURE 2. MAP OF TRANSMIGRATION VILLAGES

Each green dot is one transmigration village, 1979–1988.

The authors turned a 1998 government census into a layer of points, then matched it with climate and soil grids.

This is exactly the workflow you will learn today, just with real data and a bigger question.

Figure: Bazzi, Gaduh, Rothenberg & Wong (2016), *AER* 106(9), Figure 2. [full reference](#) →

Forced cultivation in Java: three data types in one map

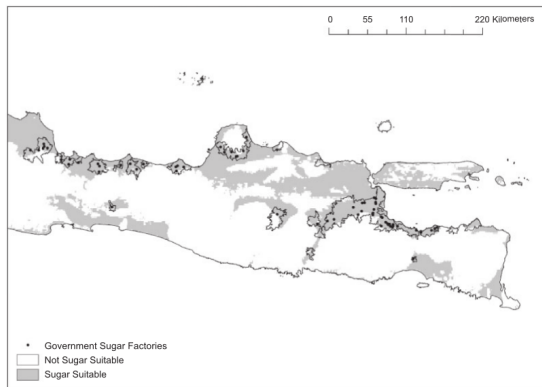


FIGURE 3
The Cultivation System.

Sources: [Commissie Umbgrove \(1858\)](#). Black dots are government sugar factories. Lines are forced cultivation catchment area boundaries. Gray areas have an elevation and slope that are suitable for sugar cultivation, whereas white areas do not.

Black dots: colonial sugar factories.
Lines: the boundary of each forced cultivation area. Grey: land suitable for sugar.

Points, lines and a grid – all three in one picture, and all three are things you will build yourself today.

Those boundaries are what the authors use to measure the long-run effect.

Figure: Dell & Olken (2020), *Review of Economic Studies* 87(1), Figure 3. [full reference](#) →

Market access: a network becomes one index



Donaldson and Hornbeck combine the railway network with the size of every neighbouring market into **one market access number per county**, then map how it changed from 1870 to 1890.

The original caption says the counties are split into seven groups with *the same number of counties* in each group.

That is **quantile** classification – the same choice we compare later today.

Figure: Donaldson & Hornbeck (2016), *QJE* 131(2), Figure III. [full reference](#) →

Spatial data in economics research

Study	Spatial data used	What it was for
Henderson, Storeygard & Weil (2012)*	Satellite night lights	A stand-in for growth where statistics are weak
Dell (2010)*	Historic <i>mita</i> boundary, Peru	Geographic discontinuity for identification
Donaldson (2018)	Colonial railway network, India	Trade costs between regions
Donaldson & Hornbeck (2016)*	US railways	Building a market access index
Bazzi et al. (2016)*	Transmigration origins and destinations	Skill transferability in Indonesia
Dell & Olken (2020)*	Forced cultivation boundaries, Java	Long-run effects of a colonial economy
Donaldson & Storeygard (2016)*	Satellite imagery across fields	A survey of satellite data in economics
Harari & Wong (2026)*	Slum upgrading boundaries, Jakarta	Long-run effects of upgrading policy

* The original figure is shown on a nearby slide. **The point:** none of them started from a map. They started from an economic question, and used maps because there was no other way to measure what they needed.

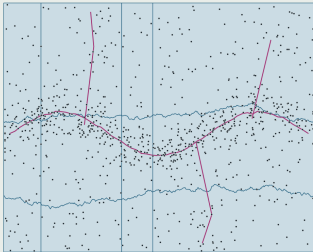
What spatial data is actually for

- 1 **It creates new variables.** Images and maps become columns of numbers: distance, area, elevation, brightness, rainfall.
- 2 **It creates identification.** Borders, coastlines and transport networks give you variation you can argue is exogenous.
- 3 **It helps you check your work.** Mapping your residuals or your treatment shows patterns a regression table hides.

Spatial Data Basics

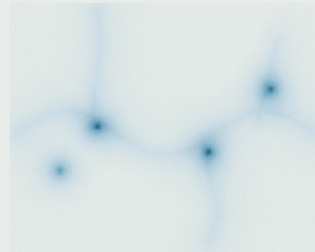
Shapes with edges, or a grid of numbers

Vector



District polygons, roads and village points: three separate layers, each with its own table.

Raster



Night lights on a grid of cells. One number per cell, nothing else.

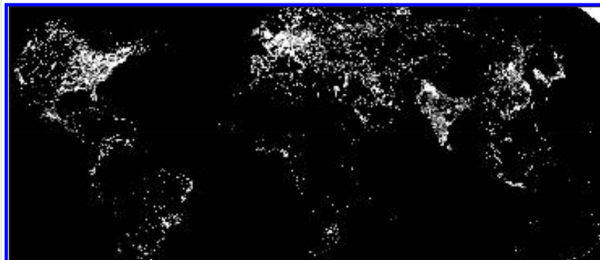
Vector: points, lines and polygons

A vector file holds coordinates and a table, nothing else. Which of the three shapes it holds decides what you can do with it.

Shape	Typical layers	What you build from it
Points	Schools, clinics, markets	Distance, density, service coverage
Lines	Roads, railways, rivers, power	Transport cost, connectivity
Polygons	Provinces down to villages	Usually the rows of your dataset

One file holds one shape only. District polygons and school points always arrive as separate layers, and you join them yourself.

Raster: the world as a grid of numbers



Resolution is how wide one cell is on the ground. A raster has no attribute table: to use it in a regression, you must summarise its values inside a boundary. That step is called *zonal statistics*.

Resolution	Example data
30 m	Forest cover, SRTM elevation
~500 m	VIIRS night lights
~5 km	CHIRPS rainfall

Figure: Henderson, Storeygard & Weil (2012), *AER* 102(2), Figure 1. [full reference](#) →

File formats you will meet

Format	Extension	What to remember
Shapefile	.shp .shx .dbf .prj	Move all the files, not just the .shp
GeoPackage	.gpkg	One file, many layers. QGIS prefers this
GeoJSON	.geojson	Plain text, easy to inspect
GeoTIFF	.tif	Raster. Stores its coordinate system inside
CSV with coordinates	.csv	Points only. Needs a latitude and a longitude column

Watch out

The .prj file holds the coordinate system. Without it, QGIS does not know which part of the planet your map belongs to.

The attribute table: your bridge to Stata and R



	fid	kode_kab	nama_kab	reformasi_izin	luas_km2	n_desa
1	1	6201	Sungai Hulu	1	6085,7	
2	2	6202	Batang Tengah	0	12584,3	
3	3	6203	Muara Baru	1	5564,8	
4	4	6204	Tanjung Lestari	0	31913	
5	5	6205	Bukit Raya	1	9484,9	
6	6	6206	Danau Panjang	0	22271,6	1
7	7	6207	Kuala Sepakat	1	7729	
8	8	6208	Rimba Jaya	0	39730,2	2

Behind every vector layer sits an ordinary table. One row is one shape on the map.

The code column is your join key. Area is computed for you.

Once geometry becomes columns of numbers, the rest is econometrics you already know.

Coordinate systems and projections

The Earth is curved and your screen is flat, so every layer carries one rule for translating between the two. Three things to keep straight:

	Example	What it is for
Geographic CRS	EPSG:4326	Latitude and longitude in degrees. Storing and sharing
Projected CRS	EPSG:327xx	Units in metres. Required before any distance, area or buffer
Datum	WGS 84 / SRGI	The reference shape of the Earth. WGS 84 is fine for most economics

Watch out

QGIS reprojects layers so they *look* right together, but it computes distance and area using each layer's own CRS. A correct picture does not mean correct numbers.

Choosing a CRS for Indonesia

What you are doing	System	EPSG
Storing and sharing data	WGS 84 geographic	4326
Distance and area in Sumatra	UTM zone 46–47	32746–32747
Distance and area in Java and Kalimantan	UTM zone 48–49	32748–32749
Bali, Nusa Tenggara, Sulawesi	UTM zone 50–51	32750–32751
Maluku and Papua	UTM zone 52–54	32752–32754

Codes starting 327xx are southern zones, 326xx are northern ones. Indonesia spans nine UTM zones, so for a national analysis either work zone by zone or use an equal-area projection.

Units of analysis and BPS codes

Level	How many	Code digits	Note
Province	38	2	Too coarse for most micro questions
District / city	± 514	4	The most common unit in published statistics
Sub-district	$\pm 7,200$	7	Fine enough for local variation, still manageable
Village	$\pm 84,000$	10	Available in the village census

Watch out

When districts split, their codes change from year to year. For a long panel, build a crosswalk table or map every year back to the boundaries of one base year.

When the unit of analysis changes the answer

Scale effect

Take the same micro data and average it up to provinces, then districts, then sub-districts. You get different coefficients. Sometimes different signs.

Zoning effect

Even with the same number of units, where you draw the lines changes the result. Administrative borders are not economic borders.

Choose your unit from the **economic mechanism you believe in** – a local labour market, a river basin, a commuting zone – not from whichever file was easiest to download.

Then report the result at another unit as a robustness check. A referee will ask.

Related trap: a relationship that holds across districts does not have to hold across the people inside them. The name for all of this is the Modifiable Areal Unit Problem (Openshaw 1984).

Where to get spatial data for Indonesia

Source	What it has	Notes
BPS statistical area maps	Provinces down to villages	Codes match published statistics. Start here
Ina-Geoportal (BIG)	Base maps and official borders	The official reference if borders are disputed
GADM	Global administrative borders	Quick for practice; not always up to date
OpenStreetMap	Roads, buildings, facilities	Coverage varies a lot between regions
VIIRS and DMSP	Night lights	A proxy for activity; be careful joining the two series
WorldPop	Population density at 100 m	A denominator for per-capita variables
CHIRPS	Daily rainfall since 1981	The usual source of weather shocks
Global Forest Change	Forest cover at 30 m	Yearly deforestation down to the pixel
SRTM / DEM	Elevation and slope	Geographic controls and instruments

The data behind these slides

What the exercises run on

District polygons, village points, a road network, a night lights raster, and one statistics table.

These files are **synthetic**. They were generated for this class so the patterns read clearly, and every place name in them is invented. Nothing here describes a real district.

Takeaway

Practising on synthetic data is fine. Reporting results from it is not. Swap in the real sources before anything goes near a thesis.

Where the real versions live

Each layer has a real counterpart on the sources slide: BPS for boundaries and statistics, BIG for official base maps, OpenStreetMap for roads, VIIRS for night lights, CHIRPS for rainfall.

Figures from published research

Every borrowed figure carries its full citation on the slide where it appears.

Spatial Operations

The operations you keep coming back to

Operation	What it does
Attribute join	Attaches a statistics table to polygons through a shared code. Exactly like <i>merge</i> in Stata
Spatial join	Attaches attributes by location: which point falls inside which polygon
Buffer	Draws a zone of fixed distance around points, lines or areas
Clip and intersect	Cuts one layer using the boundary of another
Dissolve	Merges small polygons into larger units
Zonal statistics	Reduces millions of raster cells to one number per area

Almost every variable in an applied paper is some combination of these six, run in the right coordinate system and exported to a table.

Joining by code, joining by location

Attribute join

The key is a column both tables share, usually the district code.

```
polygon.code = table.code
```

Geometry is not used at all.

Spatial join

The key is the geometry itself. No shared column needed.

```
intersects | within | contains |  
nearest
```

Use this when your data only has coordinates.

Takeaway

An example that uses both: how many clinics are within 5 km of each village? Buffer first, then spatial join with *intersects*, then count.

Zonal statistics: a raster becomes rows of data



The magenta line marks one zone.
The tool reads every cell inside it and
returns a single number for that
district.

One district

count	5,909 cells
mean	3.299
sum	19,492.7
area	29,096 km ²

Always ask for **count** too: a small district
may contain only a dozen cells.

Repeat it every year: deforestation in Riau



The same patch of Riau province in 2001 and in 2008. A yearly raster summarised by area, then stacked into a district-by-year panel.

Think about it: with eight years like this, what variable would you build?

Figure: Donaldson & Storeygard (2016), *JEP* 30(4), Figure 1 (two of eight panels); original source Burgess et al. (2012). [full reference](#) →

The standard workflow

1. **Collect your layers.** Keep the raw files in their own folder and never edit them.
2. **Put everything in one CRS.** Reproject to a metre-based system before anything involving distance or area.
3. **Build your variables.** Buffer, join, dissolve, zonal statistics. Each step produces one column with an economic meaning.
4. **Export the attribute table** to CSV, with the area code and the year included.
5. **Estimate in Stata or R.** From here on you do not need QGIS again.

Takeaway

Record steps 2 to 4 in the *Graphical Modeler*. A saved workflow can be re-run for the next year, and you can attach it as proof that your work replicates.

Hands-on QGIS

Why QGIS

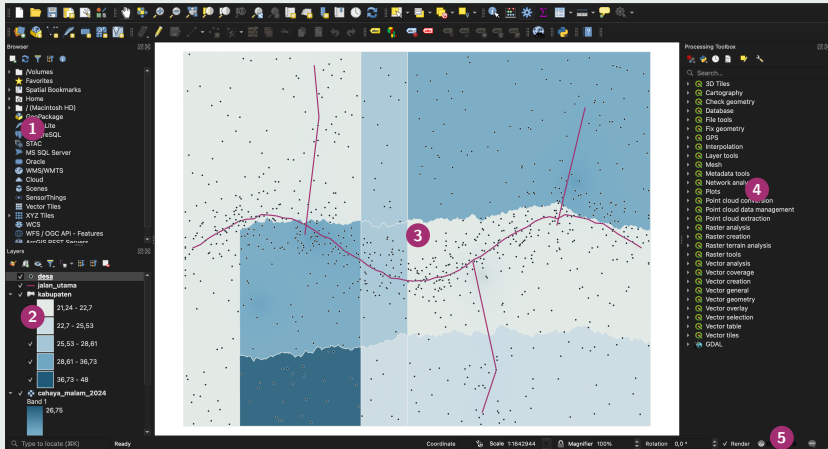
There is no licence to expire when you graduate or move institution, which also means anyone can rerun your work without buying anything.

Almost everything applied economics needs is in the standard install: joins, buffers, zonal statistics, thematic maps. Where it stops, it hands over to GDAL, GRASS, SAGA or its own Python console rather than blocking you.

The part that matters most for a thesis is quieter. The Processing Toolbox logs every step you run, and the Graphical Modeler turns a sequence of steps into a saved workflow you can run again next year, or attach as evidence that your results replicate.

Download the **LTR** release from qgis.org. The standard install is enough for today. No extra plugins.

The QGIS window, and the parts you actually use



What each part is for

1. **Browser panel.** Browse files on your computer, databases and online services without leaving QGIS.
2. **Layers panel.** The layers you have open. Top to bottom is drawing order, and the legend for each class appears here.
3. **Map canvas.** Your working view, not the final product. Maps for a thesis are made in Print Layout.
4. **Processing Toolbox.** Every analysis tool. You do not need to memorise menus: just type the tool name in the search box.
5. **Status bar.** Scale, cursor coordinates, and the project EPSG code in the bottom right corner.

Lost a panel? `View > Panels` `Toolbox: Ctrl+Alt+T`

Which file for which exercise

From billytulungen.com/lab → download. The same list sits in the README inside the zip.

Exercise	Open these
1 Open a layer, check its CRS	<code>kabupaten.gpkg</code>
2 Join a statistics table	<code>kabupaten_polos.gpkg</code> and <code>statistik_kabupaten.csv</code>
3 Thematic map	<code>kabupaten.gpkg</code>
4 Buffer and spatial join	<code>desa.gpkg</code> and <code>jalan.gpkg</code>
5 Zonal statistics	<code>nightlights.asc</code> and <code>kabupaten.gpkg</code>
6 Point layer from a CSV (take home)	<code>village_coordinates.csv</code>
7 Print Layout (take home)	the map you built in Exercise 3

Also inside: `elevasi.asc` to repeat Exercise 5 on, and `podes_sample.csv`, which joins to `desa.gpkg` on `id desa` with 25 of 888 codes unmatched.

Watch out

Everything is stored in `EPSG:4326`, degrees. Reproject to `EPSG:32749` before any step that measures.

Exercise 1: open a layer and check its CRS

1. Layer > Add Layer > Add Vector Layer, pick `kabupaten.gpkg`.
2. Right-click the layer, Properties > Information, and read the CRS line.
3. Compare it with the EPSG code in the bottom right. That is the *project* CRS, not the layer CRS.
4. Right-click the layer, Open Attribute Table (F6). Count the rows.
5. Use Identify Features, click one polygon, and read its attributes.

Open first: `kabupaten.gpkg`



The screenshot shows the QGIS interface with the 'Properties' dialog open for the 'kabupaten.gpkg' layer. The 'Information' tab is selected, displaying metadata about the layer's storage, encoding, geometry, extent, and feature count. The 'Coordinate Reference System (CRS)' tab is also visible, showing the layer's CRS as EPSG:4326 - WGS 84, which is a geographic coordinate system using degrees.

Information from provider	
Storage	GPKG
Encoding	UTF-8
Geometry	Polygon (Polygon)
Extent	111.0000000000000000, -3.5000000000000000 : 116.0000000000000000, 0.5000000000000000
Feature count	12

Coordinate Reference System (CRS)	
Name	EPSG:4326 - WGS 84
Units	Geographic (uses latitude and longitude for coordinates)
Type	Geographic (2D)
Method	Lat/long (Geodetic alias)
Celestial Body	Earth
Accuracy	Based on <i>World Geodetic System 1984 ensemble</i> (EPSG:6326), which has a limited accuracy of at best 2 meters.
Reference	Dynamic (relies on a datum which is not plate-fixed)

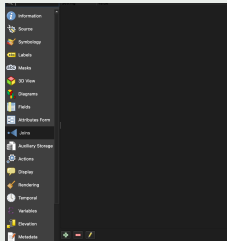
Watch out

The CRS block names the units. *Geographic* means degrees, which cannot measure distance or area. Reproject first.

Exercise 2: join a statistics table to the map

1. The key is `kode_kab` on the map but `kode` in the CSV.
2. **Layer > Add Layer > Add Delimited Text Layer**, *No geometry*. Untick *Detect field types*.
3. Right-click the polygon layer, **Properties > Joins**, press plus, and set the key field on each side.
4. Tick *Custom field name prefix* so column names do not clash.
5. Open the attribute table. All twelve rows should be filled.

Open first: `kabupaten_polos.gpkg` and `statistik_kabupaten.csv`



Takeaway

Leave *Detect field types* ticked and QGIS reads `kode` as a number while the map holds text. The join then returns twelve empty cells and no warning at all. Try it both ways.

How a join fails without telling you

Symptom	Cause	Fix
Every joined column is empty	Key types differ: text on one side, number on the other	Make them match with the Field Calculator
Some rows are empty	Codes changed when districts split	Use a crosswalk table between years
Leading zeros disappear	Excel read the code column as a number	Import the CSV with the key column forced to text
Row count grows	The key is not unique on one side	Check for duplicates before joining
Names do not match	Spelling, abbreviations and capitalisation differ	Join on codes, never on names

Watch out

A partly failed join produces no error message. The regression still runs, and the sample is no longer random. Count rows before and after.

Exercise 3: make a thematic map

1. Right-click the layer, **Properties** > **Symbology**.
2. Change *Single Symbol* to *Graduated*.
3. Set *Value* to `kemiskinan`, the poverty rate.
4. Pick a *Color ramp* and a *Mode*, then press *Classify*.
5. Set the number of classes. Five is enough for almost any map.
6. Apply, then change *Mode* and watch what happens.

Open first: `kabupaten.gpkg`



Takeaway

Step 6 is the real lesson: the same map can tell different stories.

One column of numbers, three different maps

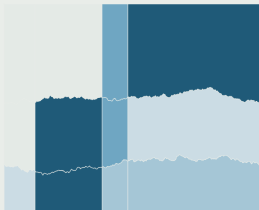
Equal Interval



26.6 · 31.9 · 37.3 · 42.7

Cuts spread evenly across the range.

Quantile



22.2 · 23.8 · 26.4 · 30.9

All four cuts sit below 31.

Natural Breaks



22.7 · 25.5 · 28.6 · 36.7

Cuts fall where the data is thin.

The same column, the same range of 21.2 to 48.0, five classes, the same colour ramp. Only the mode changed.

Think about it: which map would you pick if you wanted poverty to look evenly spread?

Choosing a classification and a colour scheme

Mode	When to use it
Equal Interval	Data spread fairly evenly
Quantile	Same number of areas in each class
Natural Breaks	A safe default for skewed data
Standard Deviation	To highlight who deviates

Sequential for a quantity that increases.

Diverging for values with a meaningful midpoint, such as positive and negative growth.

Avoid rainbow schemes. They have no natural order, and readers with red-green colour blindness cannot read them.

Watch out

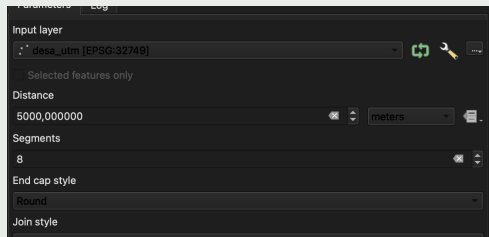
If your variable is a count, divide it by population or by area first. Otherwise your map just shows where people live, and Java will always be dark.

Exercise 4: buffer and spatial join

The question: which villages sit within 5 km of a main road?

1. Reproject both to `EPSG:32749` with `Reproject Layer`.
2. `Vector > Geoprocessing Tools > Buffer` on the roads, *Distance* 5000.
3. `Vector > Data Management > Join Attributes by Location`.
4. Villages as the base layer, the buffer as the join layer, predicate *intersects*.
5. Check your answer against `jarak_jalan_km_TRUE`, the real distance.

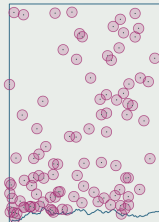
Open first: `desa.gpkg` and `jalan.gpkg`



The layer is in EPSG:32749, so the unit box beside *Distance* reads *meters*. Step 1 is what puts it there.

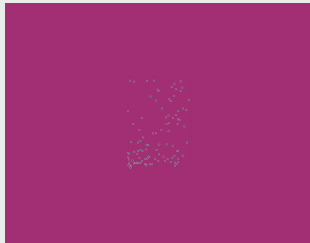
Typing 5000 in the wrong CRS

Layer in metres EPSG:32749



A 5,000 metre buffer. The circles make sense.

Layer in degrees EPSG:4326



The same number read as 5 degrees, about 555 km.

Watch out

Both of these came out of QGIS without a single error message. The one on the right still ran, still produced a new column, and could still end up in your regression.

The same buffer, continental scale

A. 40-Mile Buffers: Waterways (Black) and Railroads (Gray)



The operation you just ran in Exercise 4 is the same operation used in the *Quarterly Journal of Economics*.

Forty-mile buffers around railways (grey) and waterways (black) in 1890. Panel B of the original adds canals that were once proposed, as a comparison.

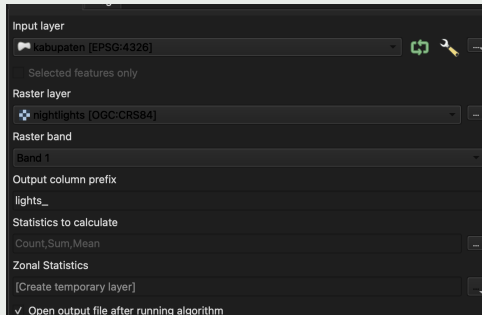
The tool is not what makes it research.
The question is.

Figure: Donaldson & Hornbeck (2016), *QJE* 131(2), Figure I. [full reference](#) →

Exercise 5: zonal statistics and export to CSV

1. Layer > Add Layer > Add Raster Layer, pick `nightlights.asc`.
2. Open the Processing Toolbox and type `Zonal statistics`.
3. Set the zones to the district polygons and the raster to the lights layer.
4. Tick the statistics you need: *mean*, *sum*, *count*.
5. Run it, then **Export > Save Features As** in CSV format.
6. Repeat for the next year and stack them into a panel.

Open first: `nightlights.asc` and `kabupaten.gpkg`



Watch out

Read the two CRS labels here. A mismatch returns an empty result.

Before you close QGIS

Save the project

Project > Save As writes a `.qgz` file holding your layers, your colours and your layout.

It does **not** copy the data. Move the project without the data folder and it opens empty.

A join is not a file

The join from Exercise 2 lives only inside the project. Close QGIS without saving and it is gone.

To keep it, run **Export > Save Features As**. That writes the joined columns into a real layer you can hand to Stata or R.

Watch out

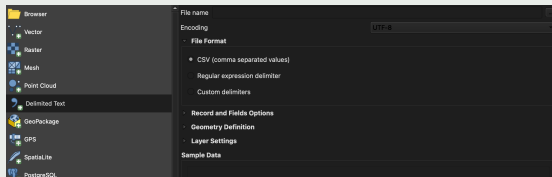
That is the live part done. The last two exercises are yours, and Exercise 7 starts from the map you made in Exercise 3. Save the project now, or you will be making that map twice.

Exercise 6 (take home): make a point layer from a CSV

1. The columns are named **X** (longitude) and **Y** (latitude).
2. **Layer > Add Layer > Add Delimited Text Layer.**
3. Choose *Point coordinates*. Set **X = longitude** and **Y = latitude**.
4. Set *Geometry CRS* to EPSG:4326 if your coordinates are in degrees.
5. Check the points land in Indonesia, then **Export > Save Features As.**

Open first: `village_coordinates.csv`

Not run in class. Do it on your own.



Watch out

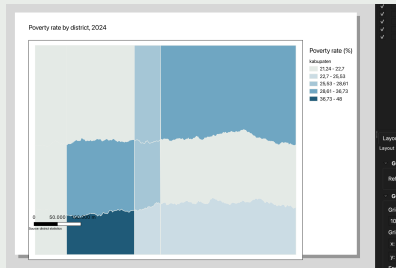
X and Y live inside *Geometry Definition*, which opens once a file is chosen. X is longitude, Y is latitude.

Exercise 7 (take home): Print Layout for your thesis map

1. **Project > New Print Layout**, name it after the figure number.
2. **Add Item > Add Map** and drag a box across the page.
3. Add a *Legend*, a *Scale Bar* and a north arrow.
4. Use *Add Label* for the title and the data source line.
5. Rename the class labels in the legend by double-clicking them.
6. **Layout > Export as PDF**, or as an image at 300 dpi.

Open first: the map you built in Exercise 3

Not run in class. Do it at home.



Takeaway

Every element appears in the Items panel on the right. A screenshot of the map canvas gives you none of this.

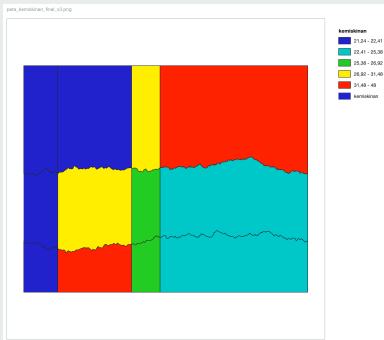
Checklist before a map leaves your desk

- The title says what, where and when.
- The legend uses clear units, not raw column names.
- There is a scale bar and a north arrow.
- The data source and year are given.
- Counts have been turned into rates.
- The classification method and number of classes are stated.
- The colours still work when printed in black and white.
- Areas with no data look different from areas with a value of zero.

The last one is missed most often, and it changes what readers conclude more often than any of the others.

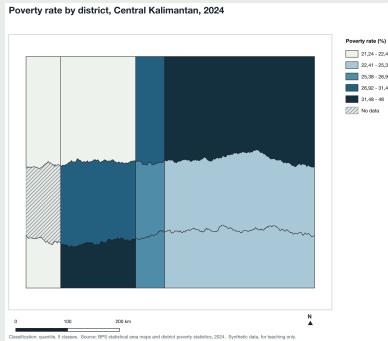
The same map, before and after that checklist

Straight out of the canvas



Rainbow scheme, raw column name, no title, no scale, no source. The no-data district is the same blue as the lowest class.

After the checklist



Same districts, same column, same breaks. Only the presentation changed, and the missing district is now visibly missing.

Back to Empirical Work

From QGIS output to a regression

$$y_{it} = \alpha + \beta \text{road_distance}_i + \gamma \text{lights}_{it} + \delta \text{rain}_{it} + \mu_i + \lambda_t + \varepsilon_{it}$$

Variable	Built with	From which layer
road_distance	Distance to Nearest Hub	Road network and village centre points
lights	Zonal statistics, mean	Night lights raster, one per year
rain	Zonal statistics, then aggregated	Daily rainfall raster
area	Field Calculator, area expression	Polygons in a metre-based CRS

Takeaway

You should be able to explain every row of this table in your defence. *Where did this number come from* is the first question an examiner asks.

Geographic discontinuity: the mita boundary in Peru

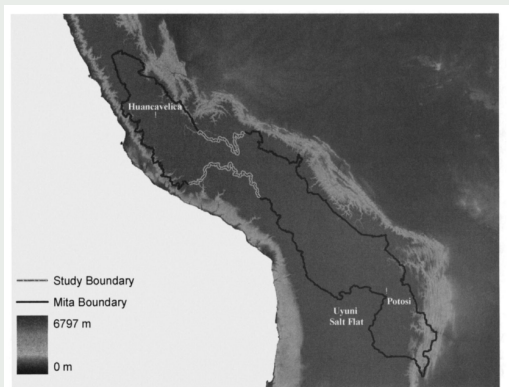


FIGURE 1.—The *mita* boundary is in black and the study boundary in light gray. Districts falling inside the contiguous area formed by the *mita* boundary contributed to the *mita*. Elevation is shown in the background.

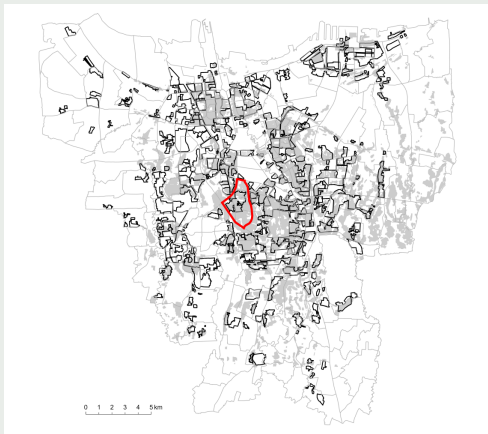
The black line is the boundary of a forced mining labour system that ran from 1573 to 1812. The background is an elevation raster.

Villages on either side of this line have similar geography, ethnicity and tax history. What differs is whether they were subject to the system.

An administrative border is treated as a treatment line – and the whole identification strategy rests on one vector layer.

Figure: Dell (2010), *Econometrica* 78(6), Figure 1. [full reference](#) →

The same idea in Jakarta



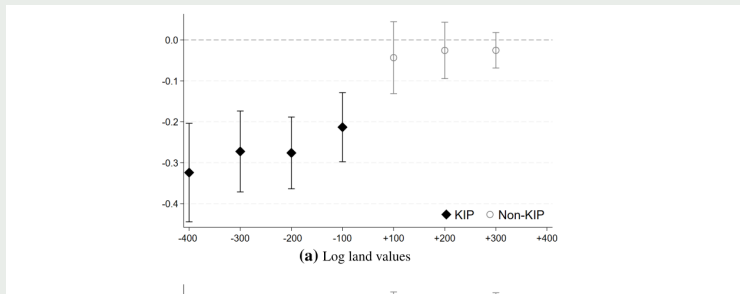
Black outlines: areas upgraded by the KIP programme between 1969 and 1984. Shaded grey: kampungs that already existed before it. The red shape in the middle is the central business district.

The programme reached 5 million people and covered a quarter of Jakarta's land.

Someone had to digitise those boundaries first. That layer of polygons is what makes the whole study possible.

Figure: Harari & Wong (2026), *Review of Economic Studies* 93(4), Figure 1. [full reference](#) →

What a distance variable looks like in the end



Land values in 100 metre bands either side of a KIP boundary. Negative is inside the upgraded area.

Think about it: that horizontal axis is a distance column computed in a GIS. This is where Exercise 4 ends up.

Figure: Harari & Wong (2026), *Review of Economic Studies* 93(4), Figure 3, panel (a). [full reference](#) →

Space as a source of identification

Strategy	What it rests on
Geographic discontinuity	Places either side of a policy border, assuming everything else changes smoothly across it
Distance as an instrument	Distance to old infrastructure or to a resource, if you can argue it reaches the outcome through one channel only
Market access index	A transport network and the size of neighbouring markets, combined into one measure of connectedness
Weather shocks	Rainfall and temperature by area and season, variation that is hard to argue people chose

Watch out

Being close together does not make something exogenous. Administrative borders are often drawn *because* of a difference: a river, a mountain, an ethnic line, or local politics.

Neighbours share shocks, so fix your standard errors

Weather, labour markets and local policy do not stop at a district border. Residuals end up correlated across space, and ordinary OLS standard errors come out too small.

1. Build a spatial weights matrix, either by adjacency or by a distance cut-off.
2. Test Moran's I on the residuals. GeoDa does it, so does the R package `spdep`.
3. If it fails, use Conley (1999) standard errors, a spatial HAC correction that lets correlation fade with distance. Stata has `acreg` and `ols_spatial_HAC`; R has `conleyreg`.

A simpler route is to cluster at a larger unit, so long as you still have enough clusters to make that honest.

Where it usually goes wrong

1. **Mixing coordinate systems.** Distances and areas become meaningless, and there is almost never an error message.
2. **A partly failed join.** Some areas drop out of the sample, the regression still runs, and the result is biased.
3. **Mapping counts instead of rates.** What you get is a population map with a different title.
4. **Reading area-level patterns as individual ones.** A relationship across districts need not hold across the people in them.
5. **A beautiful map with no question.** A map supports an argument. It is not the argument.

Practice on your own

Take district boundaries and one development indicator of your choice.

1. Join the table to the polygons by area code. Report how many rows failed to match, and why.
2. Make a five-class thematic map, and explain your choice of classification.
3. Compute the distance from each district capital to the nearest port.
4. Export the table to CSV and regress your indicator on that distance.
5. Hand in one page: map, regression table, three sentences of interpretation.

Takeaway

What is assessed is not the size of the coefficient, but whether you can explain where every number came from and see the weaknesses yourself.

Where to learn more

Learning QGIS

QGIS Training Manual
docs.qgis.org

QGIS Tutorials and Tips
qgistutorials.com

Moving to code

Geocomputation with R
r.geocompx.org

Spatial Data Science
r-spatial.org/book

Finding data

Ina-Geoportal, BIG
tanahair.indonesia.go.id

BPS statistical area maps
bps.go.id

For anything you repeat, write a script. Use QGIS to look, and R or Python to process.

References: figures reproduced in these slides

Figure numbers are links back to the slide that shows them.

- Bazzi, S., Gaduh, A., Rothenberg, A. D. & Wong, M. (2016). Skill transferability, migration, and development: evidence from population resettlement in Indonesia. *American Economic Review* 106(9), 2658–2698. [Figure 2](#).
- Dell, M. (2010). The persistent effects of Peru's mining *mita*. *Econometrica* 78(6), 1863–1903. [Figure 1](#).
- Dell, M. & Olken, B. A. (2020). The development effects of the extractive colonial economy: the Dutch cultivation system in Java. *Review of Economic Studies* 87(1), 164–203. [Figure 3](#).
- Donaldson, D. & Hornbeck, R. (2016). Railroads and American economic growth: a “market access” approach. *Quarterly Journal of Economics* 131(2), 799–858. [Figure I](#) and [Figure III](#).
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- Harari, M. & Wong, M. (2026). Slum upgrading and long-run urban development: evidence from Indonesia. *Review of Economic Studies* 93(4), 2646–2679. [Figure 1](#) and [Figure 3\(a\)](#).
- Henderson, J. V., Storeygard, A. & Weil, D. N. (2012). Measuring economic growth from outer space. *American Economic Review* 102(2), 994–1028. [Figure 1](#) and [Figure 3](#) (shown on three slides).

References: also cited

- Burgess, R., Hansen, M., Olken, B. A., Potapov, P. & Sieber, S. (2012). The political economy of deforestation in the tropics. *Quarterly Journal of Economics* 127(4), 1707–1754.
- Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics* 92(1), 1–45.
- Donaldson, D. (2018). Railroads of the Raj: estimating the impact of transportation infrastructure. *American Economic Review* 108(4–5), 899–934.
- Openshaw, S. (1984). *The modifiable areal unit problem*. CATMOG 38. Norwich: Geo Books.
- Tobler, W. R. (1970). A computer movie simulating urban growth in the Detroit region. *Economic Geography* 46, 234–240.

Software

QGIS 3.x LTR, qgis.org

Screenshots and maps in these slides were produced in QGIS on the practice data.

Figures from published papers are reproduced for teaching. Crops remove surrounding white space and panel headers only; nothing inside a panel was altered.

Data sources

Listed in full on the earlier sources slide: BPS, BIG, OpenStreetMap, VIIRS, CHIRPS, Global Forest Change, SRTM, WorldPop.

A map is not the end of your research.

It is how you build a variable
that did not exist before.

Thank you

Empirical Development Economics

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21 September 2026

billytulungen.com

These slides, the practice files and the interactive lab are all at billytulungen.com/lab.